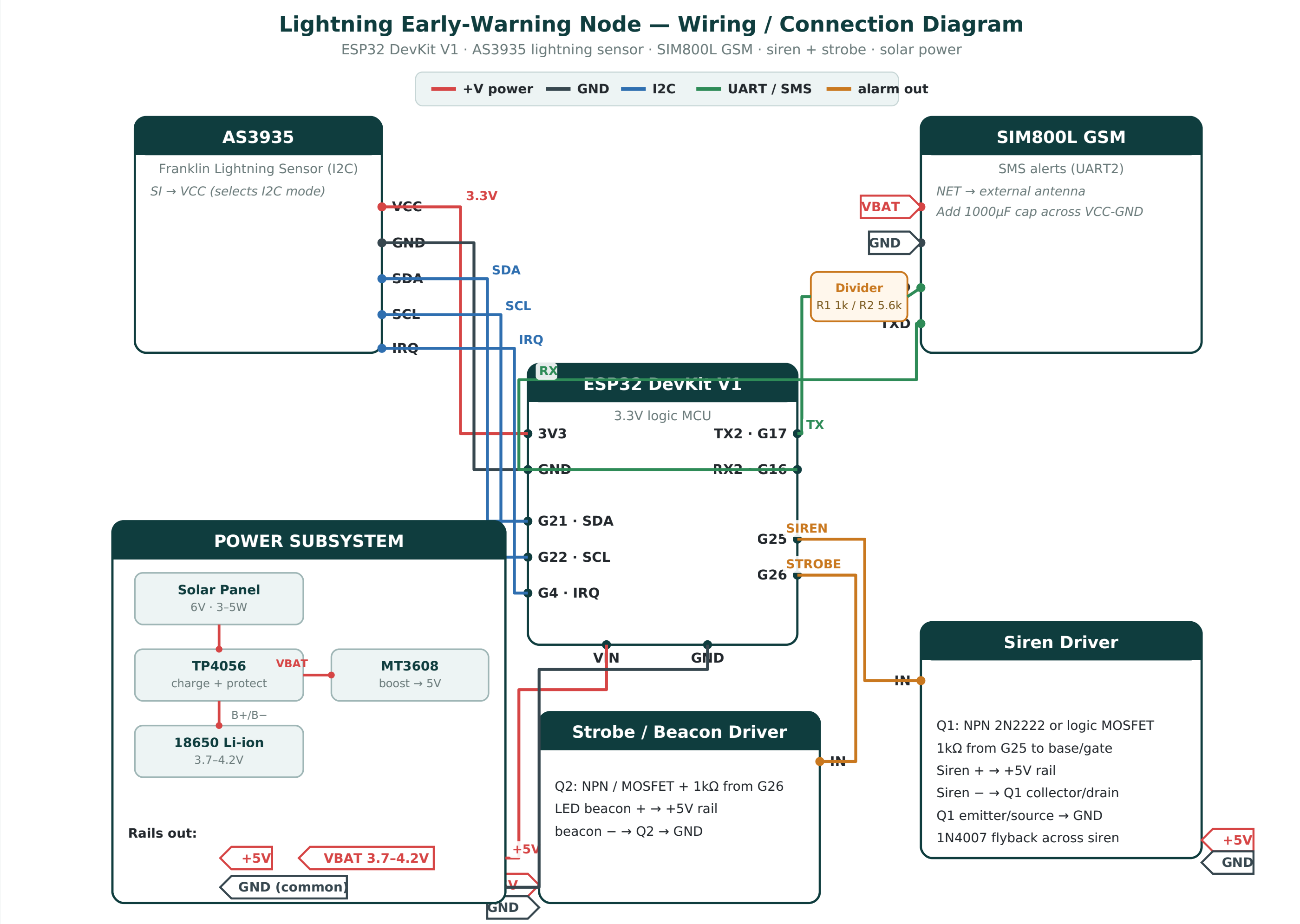
Hardware Wiring & Connection Guide

IoT-Based Lightning Early-Warning System for Field Workers

Ahoshan · ESP32 DevKit V1 · matches sketch **lightning_ews.ino** · August 11, 2026



Net labels sharing a name (+5V, VBAT, GND) are the same wire. GND is common to every module.

Pin Connections (authoritative reference)

From (module · pin)	To	Bus	Notes
AS3935 · VCC	**ESP32 · 3V3**	**3.3V**	Power the sensor from the ESP32 regulator.
AS3935 · GND	**ESP32 · GND**	**GND**	Common ground.
AS3935 · SDA	**ESP32 · GPIO21**	**I2C**	I2C data.
AS3935 · SCL	**ESP32 · GPIO22**	**I2C**	I2C clock.
AS3935 · IRQ	**ESP32 · GPIO4**	**signal**	Interrupt on lightning/disturber/noise.
AS3935 · SI	**AS3935 · VCC**	**—**	Tie SI high to select I2C mode.
SIM800L · VCC	**VBAT (18650 +)**	**VBAT**	3.7–4.2V direct from cell. Add 1000µF cap. Do NOT use ESP32 3V3.
SIM800L · GND	**GND**	**GND**	Common ground with ESP32.
SIM800L · RXD	**ESP32 · GPIO17 (TX2)**	**UART**	Through divider R1 1kΩ (series) + R2 5.6kΩ (to GND) → ~2.8V.

From (module · pin)	To	Bus	Notes
SIM800L · TXD	**ESP32 · GPIO16 (RX2)**	**UART**	Direct; 2.8V is a valid HIGH for the ESP32.
Siren driver · IN	**ESP32 · GPIO25**	**alarm**	1kΩ to transistor base/gate; flyback diode across siren.
Strobe driver · IN	**ESP32 · GPIO26**	**alarm**	1kΩ to transistor base/gate.
ESP32 · VIN	**+5V (MT3608 boost)**	**+5V**	5V from the boost converter.
ESP32 · GND	**GND**	**GND**	Common ground.

Power Wiring (build order)

- **Solar → TP4056:** panel + to TP4056 **IN+**, panel − to **IN−**. Use a 5–6V panel; a 6V panel is the practical upper limit for a basic TP4056.
- **TP4056 ↔ 18650: B+/B−** to the cell. The module's **OUT+** is your **VBAT** rail (3.7–4.2V).
- **VBAT → SIM800L VCC** directly (with a 1000µF cap). The SIM800L needs up to 2A bursts, which a boost/3.3V rail can't supply cleanly.
- **VBAT → MT3608 → +5V:** set the boost output to 5.0V, feed the ESP32 **VIN** and both alarm drivers.
- **ESP32 3V3 → AS3935 VCC.** Keep all grounds common.

Why the level shifter on SIM800L RXD

- The SIM800L RX line expects ~2.8V logic. Driving it with the ESP32's 3.3V TX can stress it over time.
- Drop it with a divider: **1kΩ in series** from GPIO17, then **5.6kΩ to GND** at the SIM RXD node (≈2.8V). SIM TXD → GPIO16 needs no shifting.

Build & safety notes

- Put a **1N4007 flyback diode** across any inductive siren, cathode to +5V, to protect the transistor.
- Mount the AS3935 **away from the SIM800L antenna, the boost converter, and switching noise**; those are classic false-alarm sources. Run the sensor in **OUTDOOR** gain (already set in the sketch).
- Calibrate the AS3935 antenna (tune-cap) once during bench testing; raise **noise floor / watchdog / spike-rejection** if you see false triggers.
- Use an **IP-rated enclosure**; keep the GSM antenna outside the metal/box and the solar panel facing the sky.
- Install the SparkFun **AS3935 Lightning Detector** library and set your phone numbers in the sketch before flashing.

The diagram is a connection schematic, not a PCB layout. Confirm your specific AS3935 and SIM800L breakout pinouts against their silkscreen before wiring.